LESSON 7.5

Represent Rational Numbers on the Coordinate Plane

Set B · Version A · Second form · supported practice

MASTERY CHECK
- ☐ Exceeds
- ☐ Meets target
- ☐ Needs practice

Name _____ **Date** _____ **Period** _____

LEARNING TARGET I can plot and identify points on the coordinate plane using ordered pairs.

WHY THIS MATTERS Captain Vega left a treasure map, but instead of an 'X marks the spot,' she hid the directions as ordered pairs! To find the treasure, you need to plot each point on the coordinate plane and connect them in order. The first clue says: Start at (2, 1), then go to (5, 4), then (8, 1).

REMEMBER

Represent Rational Numbers on the Coordinate Plane

Coordinate Plane: Plotting Points. Formula: Ordered Pair: (x, y) = (Horizontal, Vertical) An ordered pair (x, y) is two numbers that tell you exactly where a point goes. The first number moves you right or left. The second number moves you up or down.

1. Always start at the Origin (0, 0).
2. x-coordinate: Move right for positive (+), left for negative (-).
3. y-coordinate: Move up for positive (+), down for negative (-)

Words: **Coordinate plane *(Plano cartesiano)*** **Ordered pair *(Par ordenado)*** **Origin *(Origen)*** **Quadrant *(Cuadrante)*** **Reflection *(Reflexión)***

PART 1 Understand the idea

models and meaning

1 **Circle the letter of the best answer. Show how you know.**

A drone delivery company programs flight paths using coordinate pairs. A drone starts at the warehouse (0, 0), flies to pickup at (3, 6), then delivers to (8, 2), then returns to base (0, 0). In the ordered pair (3, 6), what does the 3 tell the drone?

(A) Move 3 units horizontally, along the x-axis

(B) Move 3 units vertically

(C) Fly for 3 minutes

(D) Land at altitude 3

SHOW YOUR WORK

PART 2 Apply it

real-world problems

2 **Circle the letter of the best answer. Show how you know.**

Which symbol makes this true? −6 ___ −2

(A) >

(B) =

(C) ≥

(D) <

SHOW YOUR WORK

3 **Circle the letter of the best answer. Show how you know.**

Order from least to greatest: −4, 3, −7, 0

(A) 3, 0, −4, −7

(B) −4, −7, 0, 3

(C) −7, −4, 0, 3

(D) −7, −4, 3, 0

SHOW YOUR WORK

4 **Circle the letter of the best answer. Show how you know.**

On a number line, a number to the LEFT of another number is always…

(A) greater than it

(B) equal to it

(C) farther from zero

(D) less than it

SHOW YOUR WORK

5 **Circle the letter of the best answer. Show how you know.**

Two towns record temperatures of −3 °F and −11 °F. Which town is colder?

(A) The town at −11 °F
(B) The town at −3 °F
(C) They are the same
(D) You cannot tell

SHOW YOUR WORK

6 **Circle the letter of the best answer. Show how you know.**

A drone delivery company programs flight paths using coordinate pairs. A drone starts at the warehouse (0, 0), flies to pickup at (3, 6), then delivers to (8, 2), then returns to base (0, 0). What shape does the flight path make?

(A) A square
(B) A straight line
(C) A triangle
(D) A circle

SHOW YOUR WORK

EXPLAIN YOUR THINKING

You plotted a triangle. How did the ordered pairs tell you exactly where to place each point?

SENTENCE STARTER The first number in the ordered pair tells me ___ and the second number tells me ___. The point (5, 4) means ___.

3-Point Exit Checklist: ☐ 1. Units & Labels Verified ☐ 2. Mathematical Reasoning Stated ☐ 3. Calculations Double-Checked

How did it go? ☐ 4 · I can teach it ☐ 3 · I've got it ☐ 2 · I need a hint ☐ 1 · I need help

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LEARNING TARGET I can plot and identify points on the coordinate plane using ordered pairs.

WHY THIS MATTERS Captain Vega left a treasure map, but instead of an 'X marks the spot,' she hid the directions as ordered pairs! To find the treasure, you need to plot each point on the coordinate plane and connect them in order. The first clue says: Start at (2, 1), then go to (5, 4), then (8, 1).

PART 1 Understand the idea

models and meaning

1 **Circle the letter of the best answer. Show how you know.**

Spot 2 — The map marks the Lookout Tower at (4, 9). What does the 4 tell you to do from the origin?

(A) Move 4 units to the right along the x-axis

(B) Move 4 units up along the y-axis

(C) Move 4 units to the left

(D) Stay on the y-axis at 4

SHOW YOUR WORK

2 **Write the letter of the matching item on each line.**

Spot 4 — Match each treasure landmark to its ordered pair (x, y).

_____ Sandy Dock — (A) (7, 2)

_____ Lookout Tower — (B) (4, 9)

_____ Coconut Grove — (C) (3, 8)

_____ Hidden Cave — (D) (6, 5)

_____ Fresh Spring — (E) (1, 4)

_____ Treasure Chest — (F) (9, 6)

3 **Write the name of the correct group on each line.**

Spot 5 — Sort each treasure landmark by how far RIGHT it is, ordering by its x-coordinate from closest-to-shore (smallest x) to farthest (largest x).

Groups: **Closest (smallest x)** **Middle x** **Farthest (largest x)**

_____ Fresh Spring (1, 4)

_____ Coconut Grove (3, 8)

_____ Treasure Chest (9, 6)

PART 2 Apply it

real-world problems

4 **Circle the letter of the best answer. Show how you know.**

A drone delivery company programs flight paths using coordinate pairs. A drone starts at the warehouse (0, 0), flies to pickup at (3, 6), then delivers to (8, 2), then returns to base (0, 0). Which point is farthest to the right?

(A) (3, 6)

(B) (0, 0)

(C) (8, 2)

(D) They are all the same

SHOW YOUR WORK

5 **Circle the letter of the best answer. Show how you know.**

Spot 1 — Captain Vega's first clue says the Sandy Dock is right 7 and up 2 from the origin. Which ordered pair names the Sandy Dock?

(A) (2, 7)

(B) (2, 2)

(C) (7, 2)

(D) (7, 0)

SHOW YOUR WORK

6 **Circle the letter of the best answer. Show how you know.**

Spot 3 — Vega writes: 'The Coconut Grove is 3 across and 8 up.' Which point on the map is the Coconut Grove?

(A) (0, 8)

(B) (3, 8)

(C) (3, 3)

(D) (8, 3)

SHOW YOUR WORK

EXPLAIN YOUR THINKING

You plotted a triangle. How did the ordered pairs tell you exactly where to place each point?

3-Point Exit Checklist: ☐ 1. Units & Labels Verified ☐ 2. Mathematical Reasoning Stated ☐ 3. Calculations Double-Checked

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